

ZHUORAN PAN

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Major: Computer Science and Technology

Research: AI Safety, Blockchain

GitHub: <https://github.com/Argithun>



EDUCATION

Peking University (PKU), School of Computer Science, *M.S.*

Sept 2025 - Present

Beihang University (BUAA), School of Computer Science, *B.S.*

Sept 2021 - June 2025

RESEARCH AND PROJECT EXPERIENCE

Fuzzing for BiSheng Compiler | *Core Contributor*

July 2023 - June 2024

Participated in the Beihang-Huawei Joint Lab internship, focusing on software testing for the optimization and backend modules of Huawei's "BiSheng" compiler.

- Served as the **Lead of the Optimization Module Testing**, designing over 10 fuzzing mutation strategies for various optimizations.
- Constructed mutation and differential testing modules using heuristic algorithms.
- **Identified and located 3 defects** within the compiler for the development team.

Threat Intelligence Shortlisting via Malicious Domain Ranking | *Core Contributor* Oct 2023 - July 2024

Collaborated with Tsinghua University NISL Lab and Tencent on malicious domain threat intelligence prioritization.

- Responsible for data acquisition, reproduction of ranking algorithms, and performance evaluation.
- Developed a quantitative algorithm based on **Scorecard models** to assess the maliciousness of domains.
- Paper preparation in progress for submission to **S&P 2026**.

Knowledge-Based Jailbreak Monitoring for LLMs | *Co-First Author*

Nov 2023 - Aug 2024

Joint internship with Tsinghua University KEG Lab and Zhipu AI, researching jailbreak behaviors in Large Language Models triggered by domain-specific knowledge prompts.

- Conducted large-scale measurements of jailbreak phenomena across professional fields.
- Trained a rewriter based on measurement data and mutation strategies to generate "Knowledge-Armed" prompts for safety testing.
- **Published as Co-First Author at KDD 2025**: <https://dl.acm.org/doi/10.1145/3711896.3737014>.

User-Friendly & Trustworthy Intent-Driven Web3 Framework | *First Author* Sept 2024 - Oct 2025

Designed a unified framework to translate natural language intents into on-chain operations under the "Intent-driven Execution" paradigm.

- Proposed an **intent parsing method using Domain-Specific Language (DSL) and fuzzy compilation** to decompose complex Web3 tasks into verifiable atomic execution sequences.
- Built a parallel execution layer for automated on-chain calls and designed a **Trustworthy Intent Interaction Protocol** to enhance transaction security.
- Validated the prototype in multi-protocol environments (Uniswap, Aave), achieving an **89% intent conversion success rate**.
- Preprint available at: <https://arxiv.org/abs/2603.04168>.

AlphaSpire: Agent-Driven Quant Factor Mining Framework | *Independent Developer* Sept 2025 - Jan 2026

Designed a multi-agent collaborative quantitative research framework for automated factor generation, evaluation, and selection.

- Built a **multi-agent mining architecture** that generates factor hypotheses driven by forum insights and validates them through backtesting with historical market data.
- Implemented a closed-loop workflow: factor mining → signal generation → capital allocation.

- Achieved **Gold Level** on the WorldQuant platform. Open-sourced at: <https://github.com/Argithun/AlphaSpire>.

PROFESSIONAL SKILLS

English: CET-6 Score: 613 / 720 (Advanced proficiency).

Software: CCF Certified Software Professional (CSP) in C/C++, **ranked top 0.74% nationwide**.

Programming: Proficient in C, Java, Python; familiar with Ruby, JavaScript, Verilog.

Certifications: WorldQuant Brain Consultant Certification.

AWARDS AND HONORS

Scholarships: **National Scholarship** (2021-2022, 2023-2024); Grand Prize for Academic Excellence (Twice); First Prize for Academic Competition Scholarship (Three times).

Competitions: National 2nd Prize in Computer System Development Capability Competition (Compiler Design Track) (2023); **Meritorious Winner (M)** in MCM/ICM; 2nd Prize in Provincial Mathematics Competition.

Honors: 2025 Outstanding Graduate of Beihang University.